

Database Development with PL/SQL INSY 8311



ADVENTIST UNIVERSITY
OF CENTRAL AFRICA

Instructor:

- Eric Maniraguha | ericmaniraguha2024@gmail.com | [LinkedIn Profile](#)



6h00 pm – 9h50 pm

- Monday A -G207
- Tuesday B-G204
- Wednesday E-G207
- Thursday F-G307

January 2025

Course Introduction: PL/SQL Database Development



Credits	4	 SQL*Plus
Objective	▪ Build, manage, and optimize database applications using Oracle's PL/SQL.	
What is PL/SQL?	▪ Extends SQL with procedural programming constructs (loops, conditionals, exception handling). ▪ Enables advanced data processing and manipulation.	
Key Takeaway	• Master PL/SQL fundamentals and advanced techniques to create robust database solutions.	

Course outline (1/2)

Course Outline

Introduction to SQL Command Basics (Recap)

Overview of SQL: DDL, DML, DCL, and TCL Commands

Basic SQL Commands: SELECT, INSERT, UPDATE, DELETE

Joins, Subqueries, and Aggregate Functions

Understanding Constraints and Data Integrity

Working with Views and Indexes

Introduction to GitHub

Basics of Version Control

Setting Up a GitHub Repository

Committing, Pushing, and Pulling Code

Collaborating with Others: Branching, Merging, and Pull Requests

Best Practices for Using GitHub with PL/SQL Projects

Introduction to PL/SQL

What is PL/SQL?

Advantages of PL/SQL over SQL

Overview of the Oracle Database Environment

Oracle Installation and Setup

System Requirements for Oracle Database

Downloading and Installing Oracle Database

Configuring Oracle Database (Create and Configure Database)

Introduction to Oracle SQL Developer

Troubleshooting Common Installation Issues

PL/SQL Programming Fundamentals

PL/SQL Block Structure

Variables, Constants, and Data Types

Operators and Expressions

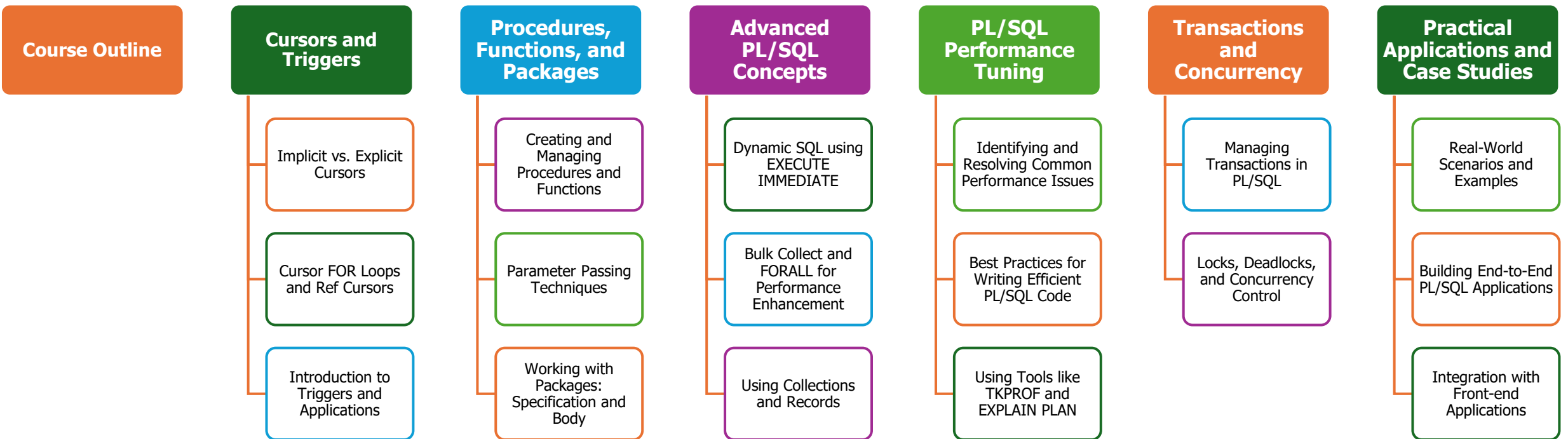
Control Structures and Error Handling

Conditional Statements (IF, CASE)

Loops (FOR, WHILE, and simple loops)

Exception Handling and Custom Exceptions

Course outline (2/2)



Course Objectives (1/2)

Master SQL and PL/SQL Fundamentals

- Review essential SQL operations including DDL, DML, DCL, and TCL to strengthen foundational database skills.
- Understand how PL/SQL enhances SQL with procedural programming features, exploring its syntax, architecture, and advantages.



Install and Set Up Oracle Database Environment

- Install Oracle Database and SQL Developer, configuring a development environment for PL/SQL programming.
- Resolve common installation and configuration issues, gaining familiarity with the Oracle database ecosystem.

Develop Core PL/SQL Blocks

- Create and manage anonymous blocks, procedures, functions, and packages to encapsulate and organize business logic.
- Apply modular programming techniques with parameter passing for building scalable and maintainable PL/SQL code.

Implement Effective Control Structures

- Use control structures such as conditional statements (IF, CASE) and loops (FOR, WHILE) to develop dynamic applications.
- Employ robust exception handling strategies, including custom exceptions, to build reliable and error-tolerant PL/SQL code.

Utilize Cursors and Triggers

- Differentiate between implicit and explicit cursors, and employ cursor FOR loops and REF cursors for effective data manipulation.
- Develop and manage triggers to automate tasks and enforce data integrity within database applications.

Create and Manage Advanced PL/SQL Components

- Implement advanced PL/SQL techniques, including dynamic SQL (EXECUTE IMMEDIATE), collections, and records to enhance code functionality.
- Design packages with a clear distinction between specification and body, optimizing code organization and reuse.

Course Objectives (2/2)



Optimize PL/SQL for Performance

- Apply performance tuning techniques like BULK COLLECT and FORALL to enhance code efficiency.
- Utilize diagnostic tools such as TKPROF and EXPLAIN PLAN to identify and rectify performance bottlenecks in PL/SQL applications.

Manage Transactions and Concurrency Control

- Control transactions using COMMIT, ROLLBACK, and SAVEPOINT to ensure data consistency and integrity.
- Address concurrency challenges, including locks and deadlocks, to manage multi-user environments effectively.

Integrate PL/SQL with SQL for Advanced Data Processing

- Seamlessly integrate PL/SQL with SQL queries to extend the functionality of traditional SQL, enabling advanced data manipulation and analysis.

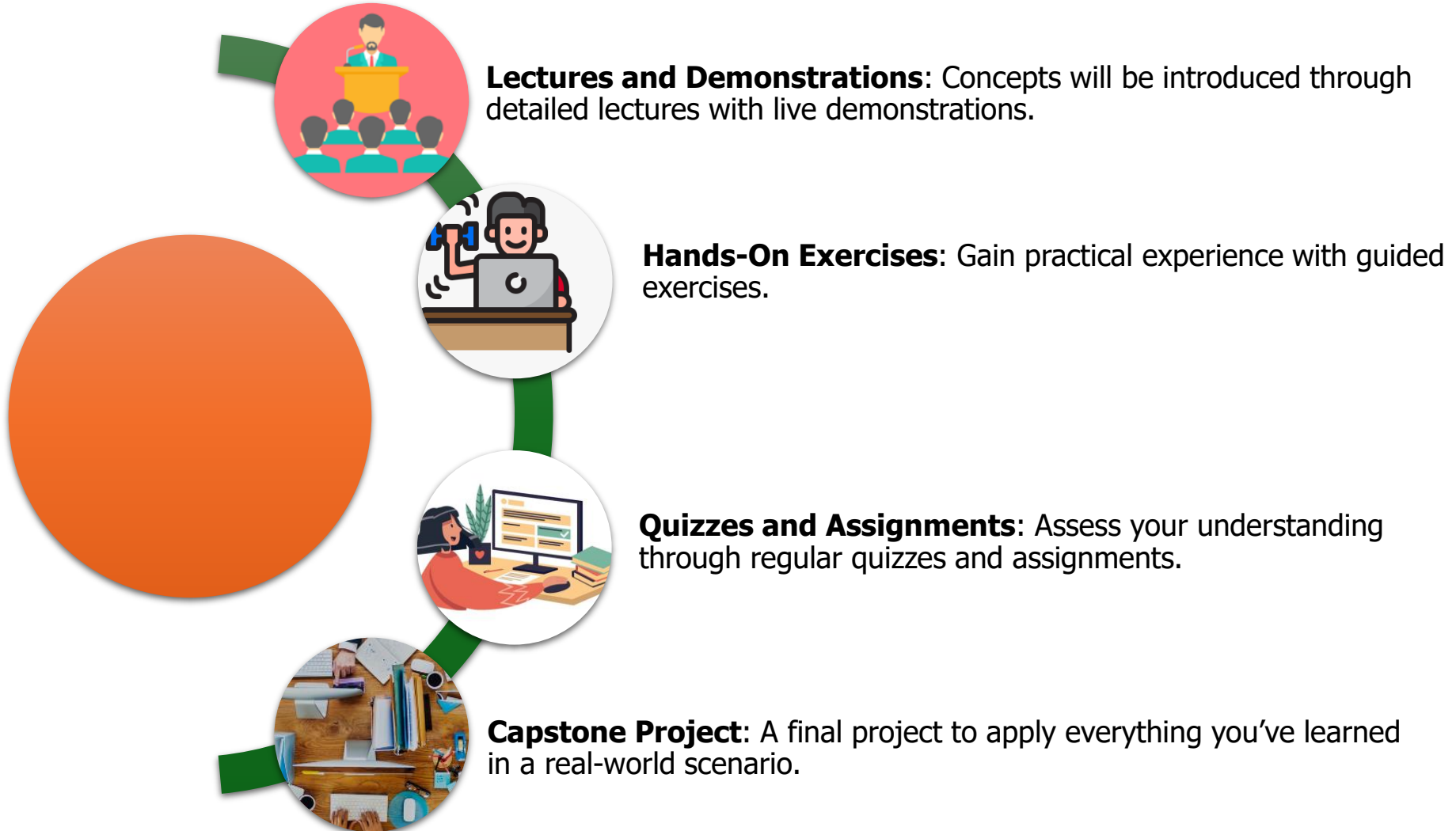
Leverage GitHub for Version Control and Team Collaboration

- Understand the basics of GitHub, including repository creation, commit management, and branch handling.
- Collaborate on PL/SQL projects using version control best practices, ensuring effective code management and teamwork.

Design and Develop Real-World PL/SQL Applications

- Apply PL/SQL programming skills to build comprehensive, real-world database applications that integrate with front-end systems.
- Demonstrate problem-solving abilities by designing, developing, and optimizing PL/SQL solutions for practical use cases.

Teaching Methodology



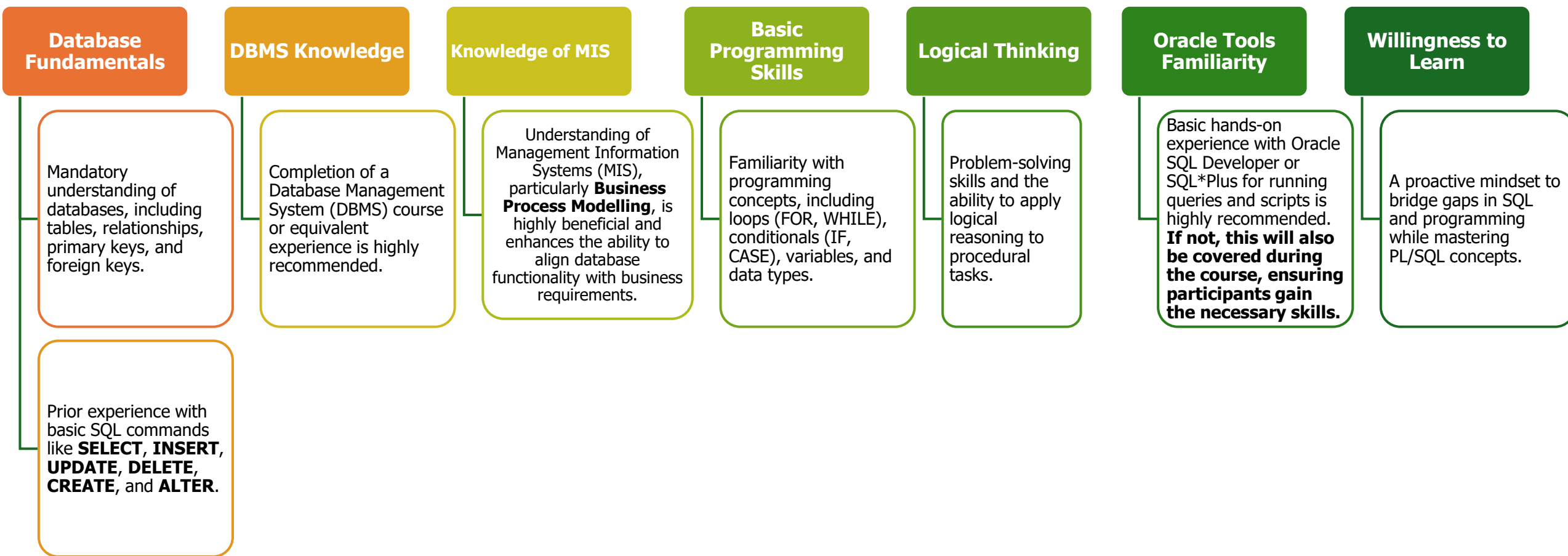


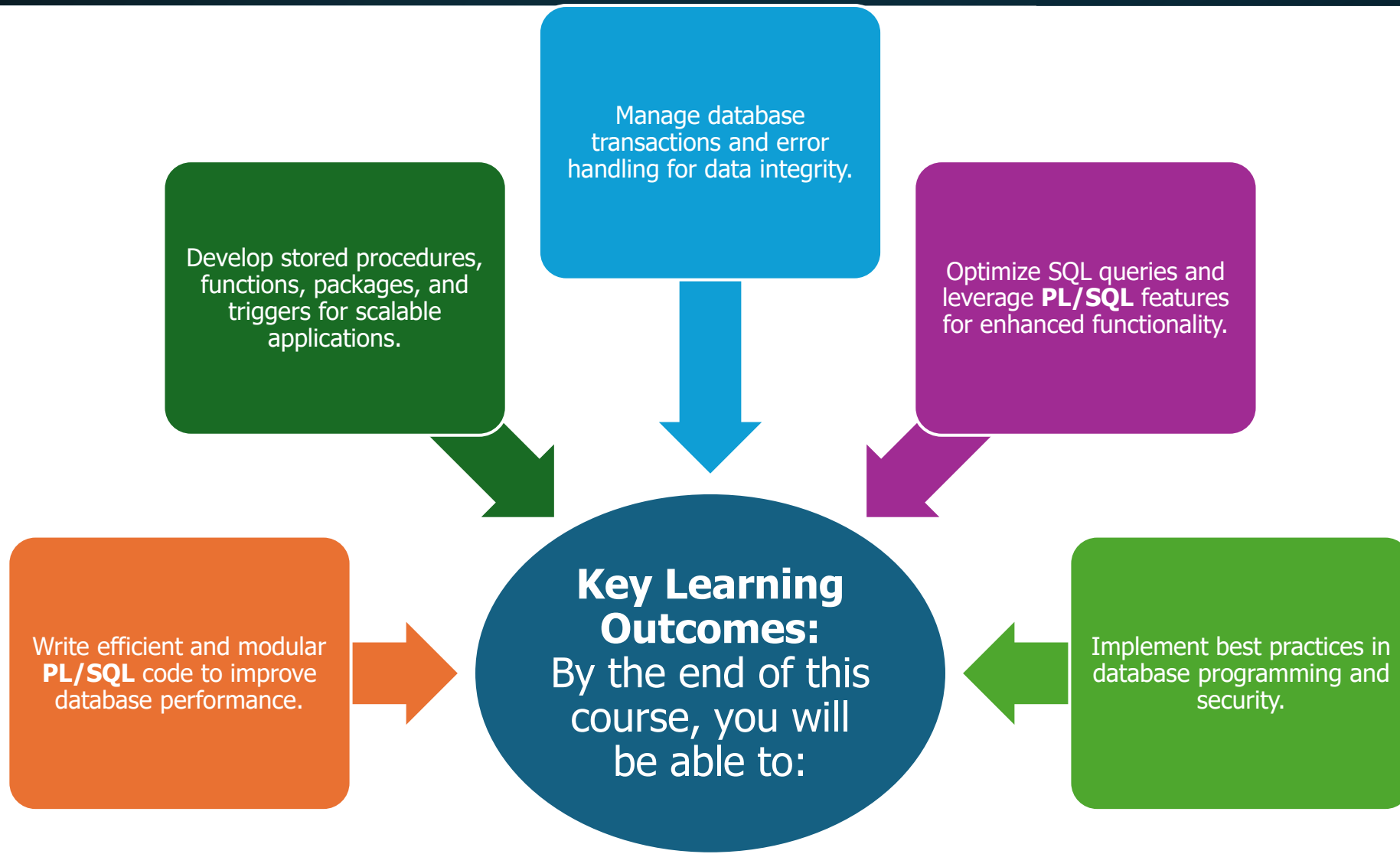
Who Should Attend?



1. Database developers and administrators.
2. SQL programmers seeking to enhance their skills.
3. IT professionals looking to specialize in **Oracle Database** development.

Prerequisites: Because Even PL/SQL Needs a Wingman (aka SQL)





Unlocking Career Paths with PL/SQL: Roles, Skills, and Future Prospects

The future after completing a PL/SQL course can be promising, especially if you're looking to build or advance a career in database management, software development, or data analysis. Here are some potential career paths and opportunities you might consider after mastering PL/SQL:



1. Database Administrator (DBA)

- **Role:** Managing and maintaining Oracle databases, ensuring their performance, security, and availability.
- **Skills Needed:** PL/SQL, database tuning, backup and recovery, user management, and SQL optimization.

2. PL/SQL Developer

- **Role:** Writing and optimizing PL/SQL code to develop backend logic, stored procedures, triggers, and functions for applications.
- **Skills Needed:** PL/SQL, SQL, performance tuning, and understanding of application development.

3. Data Analyst / Data Engineer

- **Role:** Using PL/SQL to extract, transform, and load (ETL) data, perform complex queries, and generate reports for data-driven decisions.
- **Skills Needed:** PL/SQL, data modeling, ETL processes, SQL, and data visualization tools.

4. Business Intelligence (BI) Developer

- **Role:** Designing and implementing BI solutions using PL/SQL to manipulate data for dashboards and analytical reports.
- **Skills Needed:** PL/SQL, SQL, data warehousing, reporting tools, and BI software (e.g., Oracle BI).

5. Application Developer

- **Role:** Integrating PL/SQL into applications to handle data manipulation, backend processing, and logic implementation.
- **Skills Needed:** PL/SQL, software development languages (Java, Python, etc.), and knowledge of Oracle database integration. of PL/SQL can lead to senior roles such as Lead Developer, Senior DBA, or Database Architect.

6. Oracle Consultant

- **Role:** Advising companies on best practices for Oracle database management and PL/SQL development to improve system performance and security.
- **Skills Needed:** In-depth Oracle and PL/SQL knowledge, performance tuning, and problem-solving abilities.

7. Systems Analyst

- **Role:** Analyzing and designing system processes, often using PL/SQL for data manipulation and automation within Oracle systems.
- **Skills Needed:** PL/SQL, systems analysis, understanding of business processes, and database design.



Comparison of Popular Databases: Budget, Learning Curve, Technical Expertise, and Applications



Database	Budget	Learning Curve	Technical Expertise	Applications
PostgreSQL	Open source	Moderate	Moderate	Web & enterprise applications
MySQL	Open source	Shallow	Low	Web applications
MS SQL	Commercial	Steep	High	Enterprise applications
Oracle	Commercial	Very steep	Very high	Enterprise applications

Further Learning and Certifications

1. Oracle Certification: Pursuing certifications like Oracle PL/SQL Developer Certified Associate or Professional can further validate your skills.

2. Advanced Database Technologies: Explore advanced topics like Oracle Apex, Oracle Forms, or Oracle E-Business Suite for further specialization.

3. Performance Tuning and Optimization: Gain expertise in SQL and PL/SQL tuning to become invaluable in performance-critical environments.

Why PL/SQL is Valuable:

- **High Demand:** Oracle databases are widely used in industries like finance, healthcare, telecommunications, and government, where PL/SQL skills are highly valued.
- **Versatility:** PL/SQL can be used in various roles, including application development, data management, and reporting.
- **Career Growth:** Mastery of PL/SQL can lead to senior roles such as Lead Developer, Senior DBA, or Database Architect.

4. Oracle Certification, [Click me](#)



Academic Calendar – Semester II (2024/2025)

Month	Date	Major Event
January 2025	12 th – 17 th	Orientation and Registration
	19 th	Beginning of Semester II
February 2025	1 st	<i>National Hero’s Day</i> (Public Holiday)
	5 th	Departmental Meetings (Information Management, Software Engineering, Network)
	15 th – 22 nd	Week of Prayer
March 2025	2 nd – 7 th	Mid-Semester Exams
	10 th March – 31 st May	Internship Period (12 Weeks)
April 2025	7 th – 13 th	<i>Genocide Memorial Week</i>
	25 th	End of Classes
May 2025	1 st	<i>Labor Day</i> (Public Holiday)
	27 th April – 2 nd May	Revision Week
	4 th	Final Exams Begin
	16 th	Final Exams End
	30 th	Official End of Semester II
	31 st May	End of Internship Period (12 Weeks)



Course Grading System

Assessment Component	Weight (%)	Details	
Quiz	10%	Short tests designed to assess understanding of course concepts covered in early lessons.	
Class Work (CW)	10%	In-class activities, participation, and exercises to reinforce learning and engagement.	
Assignments (Ass)	10%	Take-home tasks and practical exercises to apply concepts and demonstrate understanding.	
Mid-Semester Exam	30%	A formal assessment conducted midway through the semester to evaluate progress and learning.	
Continuous Assessment Tests (CATS)	60%	Cumulative evaluations including quizzes, assignments, and mid-semester performance.	
Final Exam (FIN EX) - Project Capstone	40%	A comprehensive exam at the end of the semester to assess overall mastery of the course.	
Total Marks	100%	Total score calculated from all assessments.	
Final Score (SCORE)	20 Points	Final grade calculated out of 20 points for academic record purposes.	15

Mastering Database Design and Development: Final Capstone Project

Final Exam (FIN EX) – 40%

- Format:** The final exam will be a capstone project designed to assess your overall mastery of the course.
- Project Scope:** Project Scope: The capstone project will incorporate all tasks covered throughout the semester, including key concepts and practices from MIS (Management Information Systems) and DBMS (Database Management Systems).
- Implementation Phases:** It will be implemented in **phases**, ensuring step-by-step progress and proper evaluation at each stage.



Why This Approach?

This structure ensures that the capstone project not only evaluates your **technical skills** but also your ability to:

- Document and communicate** findings effectively.
- Present ideas** in a professional format suitable for stakeholders and evaluators.

Key Deliverables

- 1.Problem Statement Design** – Clearly define the problem you aim to solve.
- 2.Solution Development** – Implement and test your solution.
- 3.Code Submission on GitHub** – Ensure proper version control and organization.
- 4.Documentation Report** –
 - Provide detailed explanations of the **project scope, methodology, implementation steps**, and **results**.
 - Highlight challenges faced and solutions applied during development.
- 5.PPT Presentation** –
 - Prepare a professional **PowerPoint presentation** summarizing the key aspects of your project.
 - Include sections on **problem statement, solution overview, tools used, results**, and **future improvements**.
 - Focus on visual clarity, charts, and code snippets for better understanding.



PRACTICUM

Course Policies: Attendance, Integrity, and Respect

Guideline	Details
1. Attendance Required	- Active participation is crucial for success. Regular attendance encourages collaborative learning.
	- Minimum 75% attendance required to attend the final exams.
	- Absences should be minimized and must be communicated with valid reasons.
2. Academic Integrity – No Cheating	- Always submit original work . No cheating, plagiarism, or dishonest conduct.
	- Violations will lead to disciplinary measures , including failing grades as outlined in academic policies.
3. Respect Assignment Deadlines	- Timely submission is crucial for managing academic responsibilities.
	- Late Submission Policy:
	- Assignments submitted within two weeks after the deadline will be graded at 50% of the original score .
	- Submissions later than two weeks will not be accepted and will receive zero unless prior approval is granted for valid reasons.

Course Policies: Attendance, Integrity, and Respect

Guideline	Details
4. Responsible Use of AI Tools	- AI tools like ChatGPT can support learning but should be used responsibly.
	- Ensure your work reflects your own analysis and perspective .
	- Acknowledge AI assistance when applicable.
5. Respectful Conduct	- Treat peers, instructors, and ideas with respect .
	- Engage professionally in discussions, be attentive during lectures, and contribute positively to activities.
	- Disrespectful behavior , including disruptions, will not be tolerated .
6. Start Each Class with a Prayer	- Each class will begin with a prayer to promote community and mindfulness .
	- Participation is encouraged but not mandatory . Students may observe silently if preferred.

The Power of Faithfulness: Biblical Verses and Reflections

1. Integrity in Learning and Work: Encourage students to uphold honesty and integrity in their coding practices. For example, emphasize the importance of debugging and testing code carefully and thoroughly, representing personal integrity in technical work. A relevant message could be:

"Whatever you do, work at it with all your heart, as working for the Lord, not for human masters" (Colossians 3:23). This highlights the principle of doing work with diligence and integrity, seeing each task as an opportunity to honor God.

2. Stewardship of Skills: Teach students that their skills are a gift, and they are stewards of their abilities. This perspective can help them value learning and continuously improve their skills for the good of others and society.

"Each of you should use whatever gift you have received to serve others, as faithful stewards of God's grace in its various forms" (1 Peter 4:10). This message ties technical work with a sense of responsibility to serve others and society, not just oneself.

3. Excellence in Work: Promote the pursuit of excellence in every project or task, with an emphasis on doing their best for the good of others. Excellence becomes a reflection of their commitment to personal growth and responsible use of their talents.

"Whatever your hand finds to do, do it with all your might" (Ecclesiastes 9:10). This reinforces the importance of dedication and giving one's best in everything, including technical work like programming.

4. Ethical Decision-Making: Guide students to consider the ethical implications of their work, especially in database management and handling sensitive data. Encourage them to think of their role as serving others by ensuring systems are reliable, secure, and transparent.

"Do to others as you would have them do to you" (Luke 6:31). This golden rule can be applied to ethical decision-making in the context of programming, such as ensuring privacy and fairness in database practices.

5. "Faithfulness in small things leads to greater things." — Anonymous

Thank you!

**Let's!
Start**